

Scores: What Happens to Them and Why They Matter

First, it must be emphasized that this document, and the others in Section IV of the *Judge Information Board*, are *not* essential reading for all judges. Instead, all of the documents in Section IV are intended only for those judges who want to “look under the covers” and understand how their submissions to the scoring system are converted to the form presented to the caucuses at the end of the day, and why ISEF has made the specific design decisions it has.

The Need for Scores

One obvious fact of ISEF Grand Award judging is that there are many more projects within a single subject category than any judge can possibly interview within a single day of face-to-face interviewing. The number of projects in each category typically spans from 35-85, averaging over 60, whereas the maximum number of interviews any judge can score is 17.¹ From this one fact necessarily come two results that few people appreciate before their first time serving as ISEF judges:

- Most judges will not see the best project in their category, and
- Some judges may not even see one project that is selected for any Grand Award.

Which judges these are is purely random. You are likely to be in the first group, but not the second. The reason is that before the interviews begin there is no way to predict the likelihood of any project winning an award. The algorithm that schedules all interviews is concerned solely with matching judge and student interests to the maximum extent possible and ensuring that all projects in a category get the same number of interviews (give or take one).

How, then, in their final meeting caucusing for awards, can judges decide even which projects to discuss, without everyone having an opportunity to interview every project? What is needed is a means to allow the caucus to focus its time and attention on the subset of all projects which is most likely to contain those projects that should, on the basis of the evaluations of the judges who spoke directly with them, win Grand Awards. Harsh as it may sound to the students, it does no one any good for the judges to waste valuable caucus time debating the merits of projects that are not award-worthy, when that decision has already been made by the judges who personally interviewed those projects.

The solution involves scores, a numerical ranking of projects by each judge. Scores do not determine the award-winning projects at ISEF, but they are an essential element in the process of selecting those projects.

¹ While there are 20 judging periods, instead of requiring judges to interview all day without a break (except for lunch), 5 breaks are uniformly spread throughout the first 18 periods, during which judges can choose to submit scores for up to 2 projects whose interviews were unscheduled.

Principles of Fairness in Ranking Projects

It is surprisingly easy to misuse any voting or preference scheme, inadvertently as well as deliberately, to produce results that would generally be regarded as unfair. In fact, the Arrow Impossibility Theorem shows that when comparing more than two alternatives, there is no perfect voting scheme because sensible criteria of fairness cannot all be satisfied. Thus, when selecting projects for awards at ISEF, where many more than two alternatives are in play, great care has been taken to establish a simple and straightforward process that avoids obviously unfair results.

The two “fairness” principles that have guided the design of the ISEF scoring award-selection processes are:

1. All projects must be treated equally, and
2. All judges must be treated equally.

Many simple voting systems used in various science fairs violate one or both of these principles and so can potentially produce unfair decisions.

For example, adding up the points awarded by all judges turns out to be unfair to honest judges. Even if each project would see the same number of judges,² this would be a mistake because it allows judges to engage in strategic voting. When most judges try to fairly score every project, having one judge score a single project 100 and all others zero would give that judge unfair importance in determining the winner. These are not idle speculations. This sort of scoring pattern shows up every year in real judge scores simply because every judge makes independent decisions. Every year about a quarter of all judges give scores in a narrow range of 25 or fewer points while another quarter give scores in a wider range of 50 points or more.

Rescaling scores into what are often termed “z-scores” (where the mean of each judge’s score is fixed at zero and standard deviation is fixed at unity) suffers exactly the same problem. In fact, for real scores given at ISEF, using z-scores on occasion can be worse than using raw scores.

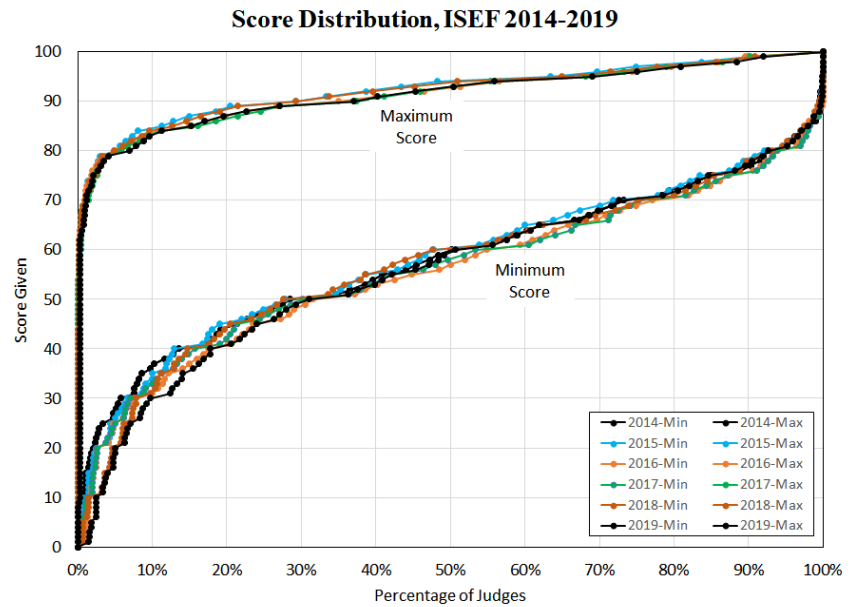
Similarly, another common system giving each judge a fixed number of tokens to use in selecting projects is also unfair to honest judges and the projects they interview. If four judges are given 10 votes to apportion between four projects, and three of the judges award identical counts of 4, 3, 2, and 1 to each of the four projects in order, then their consensus decisions would be overturned by one judge giving all 10 votes to that last project.

Many simple strategies can be misused to give some judges more power than other judges in determining award winners. Adding scores would also be unfair to certain projects based solely upon *which judges interview them*. A surprisingly robust feature of judge scoring at ISEF is this set of two facts that initially may sound contradictory:

1. 5% of all judges will not give a score below 80, and
2. 5% of all judges will not give a score above 80.

² Averaging scores over those judges that saw a given project does away with the biggest inequity, but does not avoid materially affect this issue discussed here.

The graph to the right shows the highest and lowest scores given by ISEF judges over the last 6 in-person years. The same distribution was also observed to hold among in-person judges in 2022.³ As you can see, the fraction of judges who won't give scores below 80, and those who won't give scores above 80, is somewhat larger than 5%. Regardless of the exact percentage, the point here is that some projects are going to be



interviewed by judges in the first group, and some by judges in the second. Consequently, if scores were simply added, the projects scored by judges who won't score above 80 would be at a significant disadvantage in comparison with those scored by those who won't score below 80. All projects would not be treated equally. For related reasons, the same problem arises with strategies that *linearly* transform scores into a fixed interval or use z-scores.

How ISEF Processes Scores

At ISEF, there are two separate stages in processing scores into the numeric and graphical displays judges will see in their caucus rooms at the end of the day. The first stage occurs continuously throughout the day whenever a judge submits a score and ensures equal treatment for all judges and for all projects to the extent then possible. This continues until all scores by all judges in the category are finalized and one of the category's co-chairs presses the "Begin Caucus" button, at which point the second stage is invoked. The second stage accounts for sample bias in the projects which each judge interviewed so that all projects will, in the end, be treated equally.

Stage 1: Normalization and Average Project Rank

In order to ensure that all judges and all projects are treated identically, ISEF takes all scores and normalizes them into ranks uniformly spaced within the interval $[0, 1]$. When judges give unique scores to every project, the lowest scored project is given the rank 0, the highest scored project is given the rank 1, and all other ranks are equally spaced between them. For example,

³ For unknown reasons, judges who interviewed students online in both 2021 and 2022 showed the same distribution of highest scores as in the graph, but their lowest scores were consistently higher than in-person judges. This rise was so large that the fraction of judges who did not give a single score below 80 was more than double the historical (in-person) fraction.

if there are six distinct scores, they will be normalized into the six equally spaced ranks 0, 0.2, 0.4, 0.6, 0.8, and 1.0.

In the case of ties, where a judge gives the same score to more than one project, the ranks that *would have been given* to the tied projects, had they not been tied, are averaged and assigned to all of those projects. In the example of the previous paragraph, if the six projects had only five distinct scores, with the top two projects getting the same high score, the scores would have been normalized into the ranks 0, 0.2, 0.4, 0.6, 0.9, and 0.9. The average of all normalized ranks given by a single judge, therefore, will always be 0.5, regardless of ties.

Note that this normalization process preserves only the ordering information. That is, it converts cardinal scores into ordinal ranks. A judge giving the set of scores {23, 97, 98} or the set of scores {23, 24, 98} will produce the same set of normalized ranks. Only the *relative* ranking of projects is preserved in this normalization process.

Each project can be ordered for award consideration using its *Average Project Rank*, which, as the name suggests, is the arithmetic mean of all ranks received by the project.

Stage 2: Renormalization to Account for Sample Bias

When judges are scheduled into the first 18 interview periods, nothing is known about the relative likelihood that any project will be award-worthy. Normalizing every judge's scores as described above provides an *a priori* unbiased method to compare projects to one another.

In fact, for many years, ISEF used *Average Project Rank* in displays to help caucuses focus attention on the projects most deserving of that attention. This is a good strategy, and is certainly superior to having judges nominate their favorites, since that strategy is biased toward projects seen by the more vocal judges. However, there is still one remaining source of bias contained in *Average Project Rank*.

This bias is invisible before judging begins. It becomes apparent only after the interview scores have been collected. Because judges see only a small fraction of all projects in their category, typically one-quarter, the set of each judge's projects that are normalized against each other suffers a type of sampling bias.

Simply due to the random nature of project assignments to judges, at the end of the day when all scores have been recorded it will become apparent that some judges were given slates of projects that were more award-worthy than average, and other judges were given slates that were less award-worthy. There is no way to avoid this result without knowing ahead of time which projects should win awards, which is the whole point of interviewing the projects in the first place.

The reason that this incurs a bias is easily seen by considering a judge who was given a better-than-average slate of projects. Because each judge's ranks are equally spaced from 0 to 1, the projects seen by this judge are competing for high ranks only with each other, and not with all projects in the category. On average, this judge will give those projects lower ranks than had they been interviewed by a judge who had been given an average- or lower-quality slate of projects. Conversely, judges with lower-than-average quality slates of projects give ranks to most projects that, on average, are higher than had they had a higher-quality slate of projects.

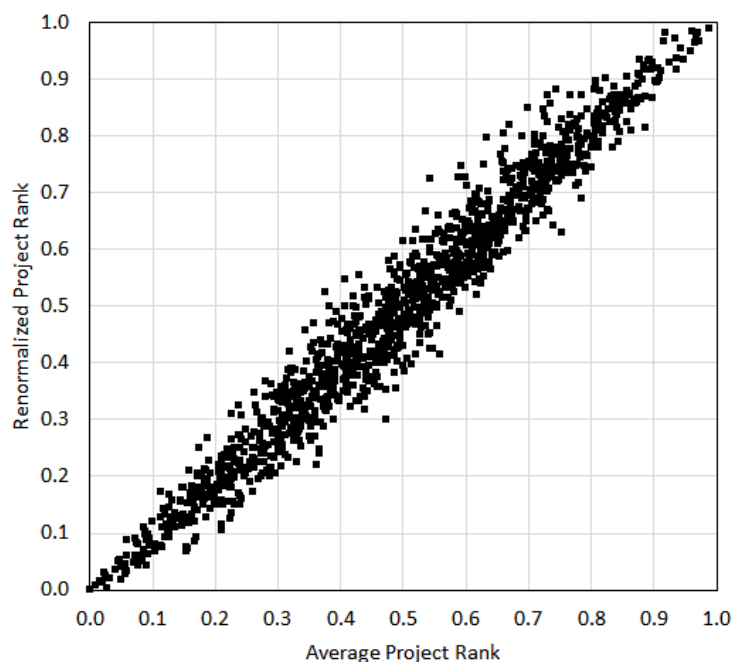
Fortunately, this bias can be corrected.

Just as fortunately, this bias does not need to be corrected until the end of the day when project rankings are presented to the caucuses, at which time the data specifying the correction are available. The correction is simple in principle, though the detailed calculation is quite laborious.

Earlier, we assured equity among judges and projects by normalizing every score by every judge into a common interval of ranks. Here, we renormalize each judge's set of ranks so that his/her *Average Project Rank* is equal to the average of project ranks determined by all judges who scored the same projects. By this renormalization we assure equity among judges and projects, despite sometimes wildly varying qualities of interview slates given to particular judges.

The renormalization is generated by a simple function that preserves each judge's top and bottom ranks at 0 and 1 (as long as those projects had those ranks and were not tied with others) and scales the interior ranks up or down by an amount that is set by the average ranks of that judge's projects determined by all judges.

Of course, renormalizing every score's rank by every judge then changes every project's *Average Project Rank* which in turn changes each judge's *Average Project Rank*, so the renormalization must be iterated. Fortunately, this process converges relatively rapidly. The final results of renormalization in ISEF 2022 for all categories are shown in the figure to the right. The change in relative project ranks by renormalization can be quite significant in some cases and is comparable for all categories. The width of the change in project ranks across all categories shown here is comparable to that in all years since ISEF began renormalizing scores in 2015.



Point of Emphasis

In order to permit judges in each category to focus attention on those projects most deserving of caucus time, every interview score is normalized to treat all projects and all judges equally, and subsequently renormalized to remove sampling bias. The relative project ranks (and rank distributions) displayed to each caucus should provide as fair and unbiased a representation of relative project quality as is possible to be deduced from scores alone.